IBM Z DATATHON 2026 · AI SECURED: INNOVATION WITHOUT EXPOSURE

# MuleNet-Z

Banks catch a third more money-mule accounts by joining their data inside an IBM Z enclave, where no bank sees another bank’s customers.

Arindam Gogoi · Niyati Joshi · Vishwajeet Sonawane · Vipul Gujar

Built and run on IBM LinuxONE

# Mule chains cross banks. Each bank sees one normal-looking hop.

THE SCAM

Stolen money hops through mule accounts at several banks within hours, then is withdrawn.

THE BLIND SPOT

Each bank sees only the transfers that touch its own customers. The chain is invisible.

THE DEADLOCK

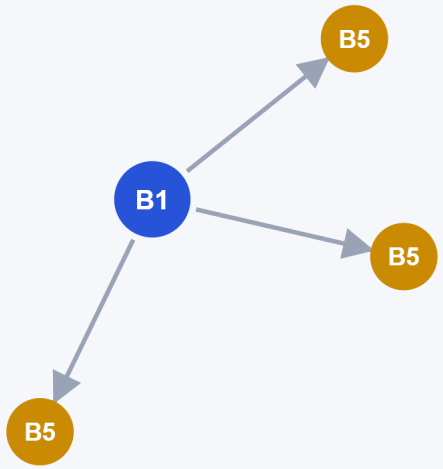
Banks cannot pool customer data: India’s DPDP Act and banking secrecy forbid it.

RBI’s MuleHunter.AI helps each bank find mules in its own books. The gaps **between** banks stay dark. That is the gap MuleNet-Z closes.

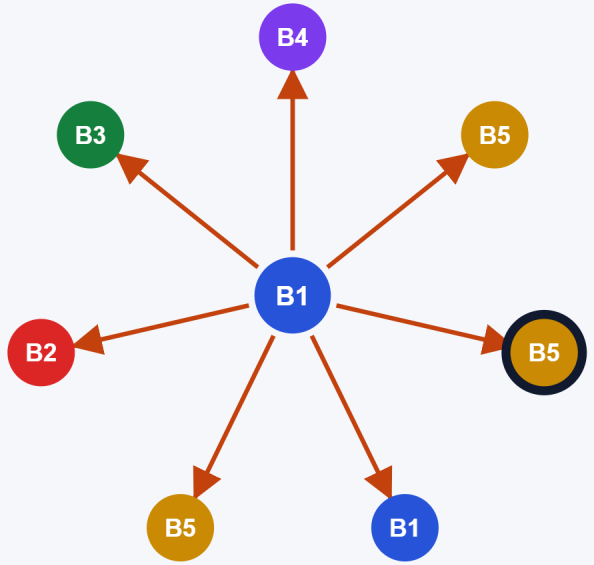
Source: RBI MuleHunter.AI coverage, Business Standard, Dec 2024

# B1 knew. B5 couldn’t find out.

**What B5 sees alone: 3 incoming transfers, no alert**




**Joined in the enclave: 7 transfers to 5 banks in 61 h, alert raised**




The sender is already labelled a mule at B1. B5’s customer received what looked like an ordinary transfer. Only the joined view connects the two, without either bank seeing the other’s data.

A real laundering pattern from IBM’s AML dataset (HI-Small). Dark ring: account alerted at the 1% budget.

# Join the data where no one can look: an attested enclave on IBM Z

**1 · Each bank**

Pseudonymises account IDs (HMAC) and encrypts its transfers with its own AES-256-GCM key.

→

**2 · Attestation**

Banks release keys only to an enclave whose code measurement they approved.

→

**3 · Inside the enclave**

One joined graph. IBM Snap ML mines laundering patterns; XGBoost scores every account.

→

**4 · Back to each bank**

An encrypted file with alerts on its own accounts only, each with reasons.

No bank, operator or administrator sees another bank’s data. Each bank alerts on its top 1% of accounts, the same workload as today.

RESULT · FULL DATA · SAME 1% ALERT BUDGET

# Joining catches 32% more mules

19.0% → 25.0% of mule accounts caught

**+6.0 pts**

95% CI +3.9 to +8.0; +5.0 to +7.5 across 4 training seeds

**160 → 193**

cross-bank laundering chains caught, of 662

**0.168 → 0.261**

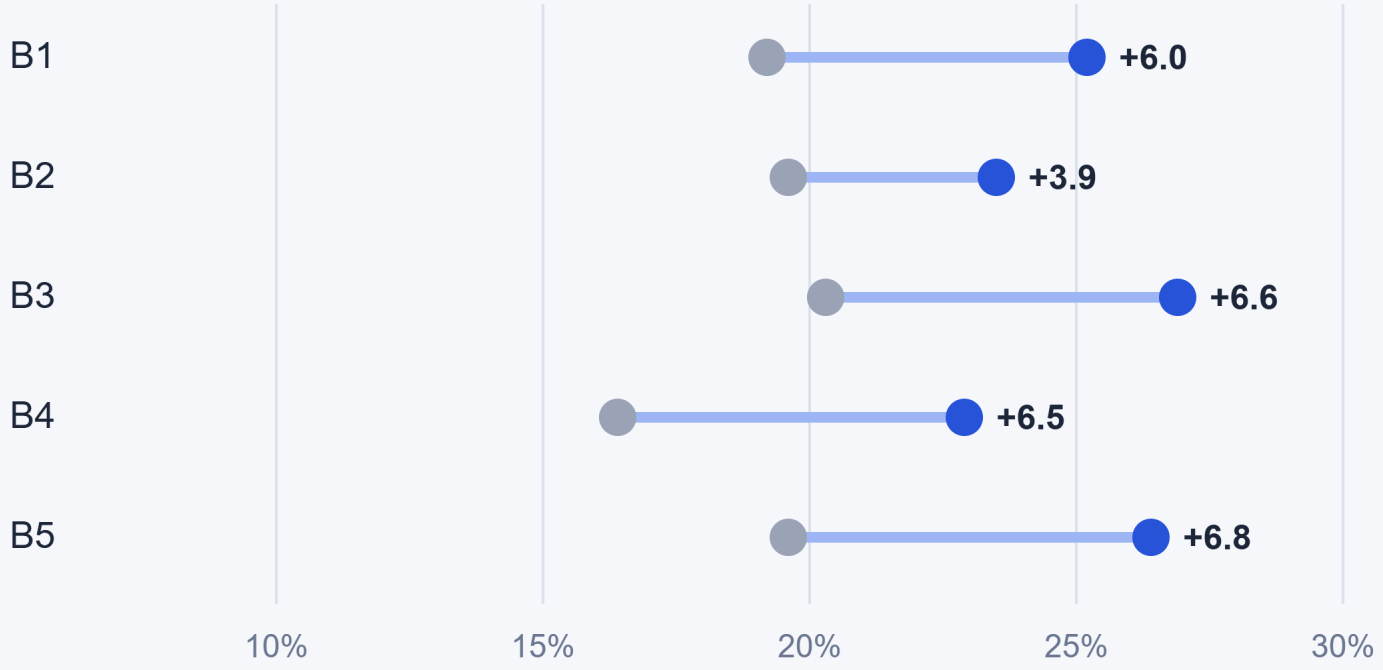
PR-AUC: the whole risk ranking gets better

**1 in 3**

alerts is a confirmed mule (30.9% precision)

IBM AML HI-Small: 5,078,345 transactions, 515,088 accounts, 5 consortium banks. Snap ML graph patterns + XGBoost, each bank alone vs joined.

# Every bank catches more mules when joined



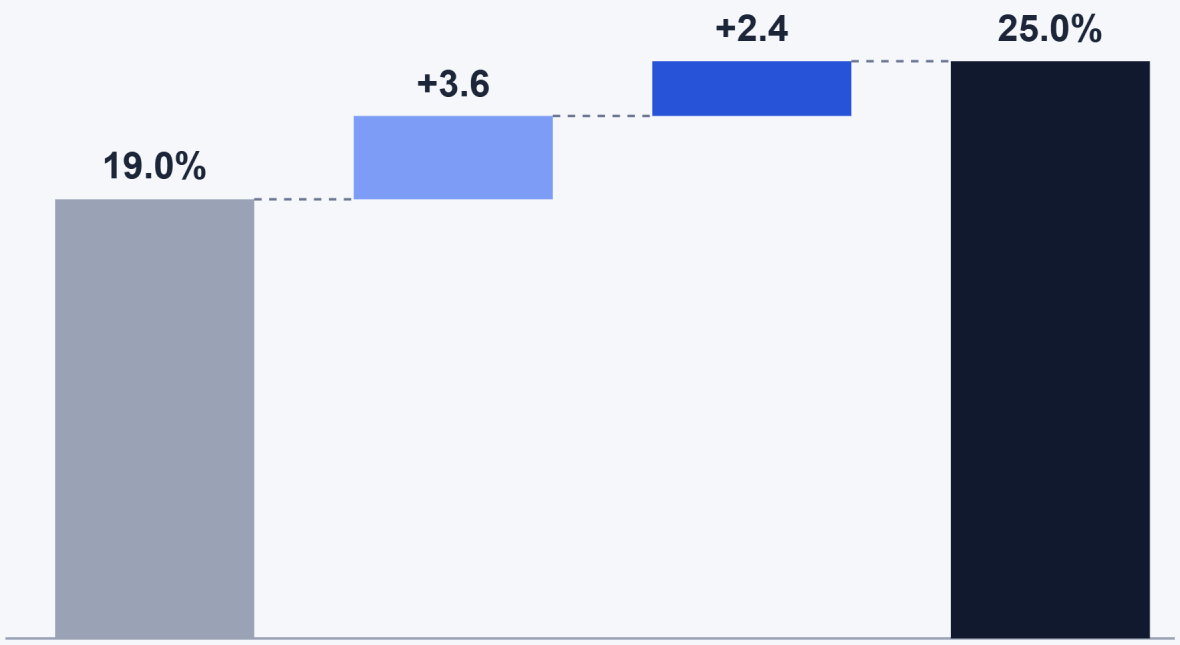

- each bank alone
- joined in the enclave

Recall at the same 1% alert budget. Even the bank that gains least, B2, gains almost 4 points.

**Second dataset** (LI-Small, 6.9M transactions, rarer laundering): 9.3% → 12.4%, +3.1 pts (CI +0.9 to +5.2). Again about a third more. Four of five banks gain there.

HI-Small test set, 1,272 mules. Per-bank results in results/metrics_per_bank.csv.

# Where the gain comes from: shared intelligence first




**Each bank alone**

its own data and labels

**+3.6 pooled labels**

what banks know, shared; federated learning can reach this

**+2.4 joined graph**

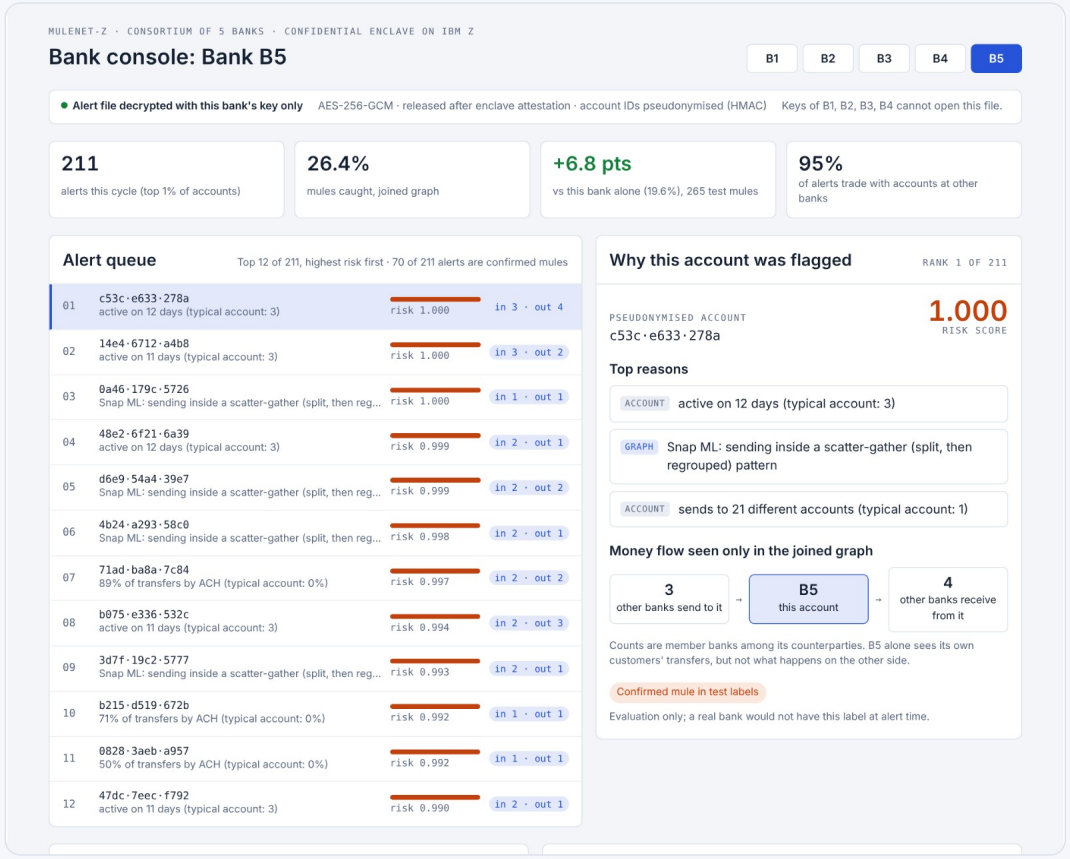
transfers no single bank sees; only the enclave adds this. Positive on all 4 training seeds (+1.0 to +2.4)

**Joined in the enclave**

LI-Small: 9.3% → +2.8 → +0.3 → 12.4%

Federated-learning proxy: one model on all banks' labels, each account's features from its own bank only. Pre-set rule (non-overlapping CIs): matches FL.

# Every alert comes with checkable reasons




**37%**

of the model's reasoning comes from IBM Snap ML graph patterns: fan-in, fan-out, scatter-gather.

**96%**

of alerts trade with accounts at other banks, evidence only the joined graph can show.

Reasons are exact per-feature contributions (TreeSHAP), written against a typical account: "receives from 9 different accounts (typical: 2)".

Live demo page: the MuleNet-Z bank console. IDs are pseudonyms; banks B1–B5 are built from IBM's synthetic AML data.

# Built and run end to end on IBM LinuxONE

## 27 min

Full experiment on 5M transactions (joined, five banks alone, ablation) on 3 vCPUs and 8 GB, streaming transactions through IBM Snap ML.

## < 3 ms

per new payment to update the graph patterns and score it, even at p99; about 45,000 payments per second.

## 6 s

for the enclave demo to decrypt five banks' submissions, join 1.1M transactions and write per-bank encrypted alerts.

## On-chip AI

Scored on the IBM Z on-chip AI accelerator via zDNN: identical alerts to the CPU. For this small model the CPU is faster (0.6 s vs 5.2 s).

IBM LinuxONE Community Cloud (machine type 9175), Ubuntu 24.04 s390x. Accelerator test: a Snap ML boosting model on the same features, zDNN 1.1.2.

# Each prototype step maps to IBM Z

| Our prototype step | IBM Z component | What it guarantees |
|---|---|---|
| Enclave code runs as a process | IBM Secure Execution for Linux | Hypervisor and admins cannot read the guest’s memory |
| Enclave image built for one server | Image encrypted with the host key document | The image boots only on that IBM Z machine |
| Hash of code and model | Secure Execution attestation | Banks verify exactly which code runs before trusting it |
| Keys released after the hash matches | Attestation-gated secrets (verify tool per generation) | Keys reach only the approved image, never the operator |
| Consortium deploys the enclave | Hyper Protect Virtual Servers, encrypted contract | No single party can change what runs or log in to inspect it |
| Bank keys and HMAC key in files | Crypto Express HSM (EP11) | Keys are used inside tamper-resistant hardware, never in plain form |

Prototype simulates these steps in software; proven: other banks’ keys cannot open a bank’s alert file. Verify component details with IBM docs.

# What we have not proven yet, and what comes next

## Limits

- The enclave boundary is simulated in software.
- IBM’s AML data is synthetic; real-bank validation is still needed.
- Accounts are split at random, not by time.
- Joining loses 39 chains that banks catch alone, while it rescues 72.
- By our pre-set rule, the gain matches a federated-learning proxy.

## Next steps

- Deploy the enclave on IBM Secure Execution with attestation.
- Hold keys in Crypto Express HSMs.
- Score on the Telum II AI accelerator with Snap ML.
- Test on a time-based split and real consortium data.
- Add audit logs and submission checks against probing.

MULENET-Z

# One join. No exposure.
# A third more mules caught.

Banks pool what they know inside an IBM Z enclave. Each gets back only its own alerts, with reasons.

Code and results: github.com/leo-leo-691/mulenet-z

Bank console demo: leo-leo-691.github.io/mulenet-z/demo

Arindam Gogoi · Niyati Joshi · Vishwajeet Sonawane · Vipul Gujar